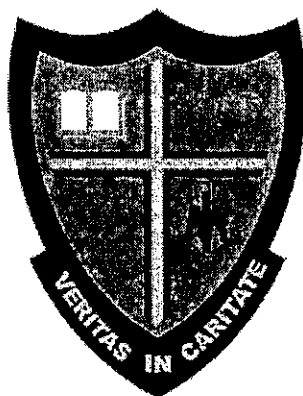




ST BENEDICT'S

SUBJECT	Mathematics	PAPER	Algebra Part B
GRADE	11	DATE	14 February 2019
EXAMINER	Miss A Balzer	MARKS	35
NAME		MODERATOR	Gr 11 Teachers
TEACHER	MEMO	DURATION	40 min

QUESTION NO	DESCRIPTION	MAXIMUM MARK	ACTUAL MARK
1	Simultaneous Equation	6	
2	Completing the Square	6	
3	Nature of Roots	23	
TOTAL		35	

QUESTION 1**6 MARKS**Solve for x and y if

(6)

$$x + 2y - 3 = 0$$

$$x^2 - y^2 - x - y = 0$$

$$x = 3 - 2y \checkmark$$

$$\therefore (3 - 2y)^2 - y^2 - (3 - 2y) - y = 0 \checkmark_{\text{sub.}}$$

$$9 - 12y + 4y^2 - y^2 - 3 + 2y - y = 0 \checkmark$$

$$3y^2 - 11y + 6 = 0$$

$$(y - 3)(3y - 2) = 0 \checkmark$$

$$y = 3 \quad \text{or} \quad y = \frac{2}{3} \checkmark \text{ answers}$$

$$\begin{aligned} x &= 3 - 2(3) \\ &= -3 \end{aligned}$$

$$\begin{aligned} x &= 3 - 2\left(\frac{2}{3}\right) \\ &= \frac{5}{3} \checkmark \text{ answers} \end{aligned}$$

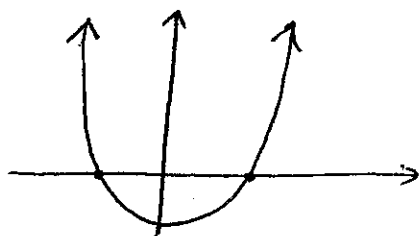
QUESTION 3

23 MARKS

1. Consider the quadratic expression $ax^2 + bx + c$. Draw a rough sketch of a parabola if the following is true for the expression.

a. $\Delta > 0; a > 0$

(2)

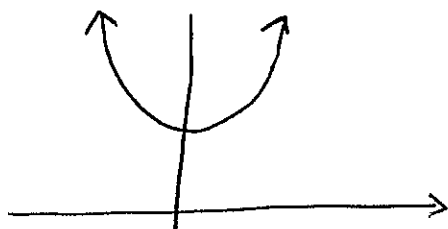


✓ direction

✓ two roots

b. $\Delta < 0; a > 0$

(2)

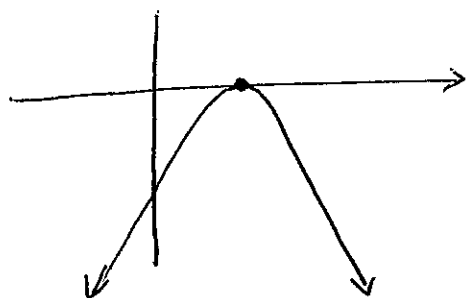


✓ direction

✓ no roots

c. $\Delta = 0; a < 0$

(2)



✓ direction

✓ one root

2. ai) Calculate Δ for $x^2 = x + 30$

(3)

$$x^2 - x - 30 = 0 \quad \checkmark$$

$$\Delta = b^2 - 4ac$$

$$= (-1)^2 - 4(1)(-30) \quad \checkmark_{\text{sub.}}$$

$$= 1 + 120$$

$$= 121 \quad \checkmark$$

ii) Hence describe the nature of the roots.

(3)

$\Delta > 0$ and perfect square

$\therefore$ Real, Rational, unequal.